

CS 3793 In-Class Quiz 2

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Question 1 (10 points):

The 6-Queens problem involves placing 6 queens on a chessboard such that no two queens attack each other. A common cost function is h , the number of pairs of queens that are attacking each other.

Suppose you use Steepest Ascent Hill Climbing without sideways moves.

You start with a randomly generated board configuration (only one queen per column).

Each successor state is generated by moving one queen within its column to a different row.

a) Explain why the algorithm might fail to find a solution.

It may fail to find a solution because it can get stuck to a local minimum (which is a state with $h > 0$ where all the possible moves leave h unchanged or make it worse). It could also be stuck on a plateau (equal h values). Since sideways moves aren't allowed, it cannot escape these and just stops before finding a solution.

b) What is the total number of states?

Total states = $6^6 = 46,656$ states

c) What is the total number of successor states for each state?

The total number of successors for each state is 30.

d) Suppose you use random restart hill climbing to improve performance in this scenario. What is the max number of restarts if the success probability $p = 0.2$?

It would be 5 restarts since $1/p = 1.0/0.2 = 5$ restarts.

e) What are the minimum and maximum values for h ?

The minimum value for h is 0. The maximum is 15.

Question 2 (10 points):

Consider using a Genetic Algorithm (GA) to solve the 6-Queens problem. In this representation, each individual (chromosome) is a 6-tuple where the i -th element indicates the row position of the queen in column i .

a) What is the role of the fitness function in a GA, and what would be a suitable fitness function for the 6-Queens problem?

The fitness function is used to measure how good a solution is. It helps to guide the choosing of which individuals to reproduce. A suitable fitness function for this problem could be the number of the non-attacking pairs of queens: $Fitness = 15 - h$, where h is the number of the attacking pairs

b) What is the range of fitness function for the 6-Queens problem?

The range of the fitness function is 0 to 15.

c) If g is a goal state for the 6-Queens problem, what is $Fitness(g)$?

$Fitness(g) = 15$

d) If s is a state that all queens are in the same row, what is $Fitness(s)$?

If they are all in the same row then every pair attacks each other, so $h=15$, meaning $Fitness(s) = 0$.

e) If s' is a state represented by the following chromosome, what is $\text{Fitness}(s')$?

$$s' = 111116$$

The first five queens (all in row 1) will form $C(5,2) = 10$ attacking pairs and the 6th queen (in row 6) will diagonally attack only the 1st queen, which adds 1 more pair, so $h = 11$ and $\text{Fitness}(s') = 15 - h = 15 - 11 = 4$.